

Home Lab 2: The Powder Workflow

Do this with a parent. Goggles on. Never taste anything.

Clean up after. See the [Home Lab Safety Agreement](#).

Goal

Practice running the same tests in the same order on four known powders, so it becomes automatic.

You need

From the kit: **baking soda, corn starch, table salt, sugar**; iodine dropper; cabbage indicator. From home: white vinegar, water, a muffin tin or 4 cups, toothpicks, paper towels.

Steps

For **each** of the four powders, in this order:

1. **Look** — write down what it looks like (crystal? fine powder?).
2. **Feel** — rub a pinch between your fingers. Gritty? Silky? Soft?
3. **Water** — stir a pinch into a little water. Does it fully dissolve, stay cloudy, or make "oobleck"?
4. **pH** — add 2 drops of cabbage indicator to the water mix. Color? Acid, neutral, or base?
5. **Acid** — put a fresh pinch on a paper towel, add a few drops of vinegar. Fizz or no fizz?
6. **Iodine** — put a fresh **dry** pinch on the paper towel, add **one** drop of iodine. Blue-black, or stays orange-brown?

Do all four. Then do it a second time, faster, and see if you can do it from memory without checking the steps.

Results

Powder	Look	Feel	In water	pH (cabbage)	+ Vinegar	+ Iodine
Baking soda						
Corn starch						
Table salt						
Sugar						

Follow-up

1. Which powder fizzed with vinegar? What gas made the fizz?
2. Which powder turned iodine blue-black? What does that tell you is in it?
3. Which powder(s) had a basic pH?
4. Give the chemical formula for each of the four powders.

Coach notes

- Baking soda: soft, dissolves, basic (blue-green), **fizzes**, no iodine change. NaHCO_3 .
- Corn starch: silky/squeaky, "oobleck," neutral, no fizz, **blue-black iodine**. $(\text{C}_6\text{H}_{10}\text{O}_5)_n$.
- Table salt: cubic, dissolves, neutral, no fizz, no iodine change. NaCl .
- Sugar: glassy grains, dissolves, neutral, no fizz, no iodine change. $\text{C}_{12}\text{H}_{22}\text{O}_{11}$.
- Follow-up: (1) baking soda; carbon dioxide. (2) corn starch; starch. (3) baking soda. (4) as above.